

CHAPTER : 2

Introduction

The entities required for the Vilvah system are identified based on the business requirements of the organization. The attributes of each entity, along with their primary keys, foreign keys, and data types, are defined to ensure proper data organization and efficient data management. The relationships between the entities are identified to represent the interactions within the system, including customer management, product management, supplier management, inventory management, order processing, and payment management. In addition, the cardinality of each relationship is determined to define how the entities are associated with one another. This analysis provides a clear understanding of the database requirements and establishes a strong foundation for designing an efficient relational database while ensuring data integrity, consistency, and minimal redundancy.

2.0 Customer

Attribute	Data Type	Key	Description
Customer_ID	INT	PK	Unique customer ID
First_Name	VARCHAR(50)	-	Customer first name
Last_Name	VARCHAR(50)	-	Customer last name
Gender	VARCHAR(10)	-	Gender
Date_of_Birth	DATE	-	Date of birth
Email	VARCHAR(100)	-	Email address
Phone	VARCHAR(15)	-	Mobile number
Address	TEXT	-	Customer address
City	VARCHAR(50)	-	City
State	VARCHAR(50)	-	State
Pincode	INT	-	Postal code
Registration_Date	DATE	-	Registration date

2.1 Product

Attribute	Data Type	Key	Description
Product_ID	INT	PK	Unique product ID
Product_Name	VARCHAR(100)	-	Product name
Category_ID	INT	FK	References Category
Supplier_ID	INT	FK	References Supplier
Price	DECIMAL(10,2)	-	Product price
Weight	VARCHAR(20)	-	Product weight
Ingredients	TEXT	-	Ingredients used
Skin_Type	VARCHAR(30)	-	Suitable skin type
Brand	VARCHAR(50)	-	Brand name
Expiry_Date	DATE	-	Expiry date

2.2 Category

Attribute	Data Type	Key	Description
Category_ID	INT	PK	Unique category ID
Category_Name	VARCHAR(50)	-	Category name
Category_Description	TEXT	-	Category description
Status	VARCHAR(20)	-	Active / Inactive

2.3 Supplier

Attribute	Data Type	Key	Description
Supplier_ID	INT	PK	Unique supplier ID
Supplier_Name	VARCHAR(100)	-	Supplier name
Contact_Person	VARCHAR(50)	-	Contact person
Phone	VARCHAR(15)	-	Phone number
Email	VARCHAR(100)	-	Email address
City	VARCHAR(50)	-	Supplier city
GST_Number	VARCHAR(20)	-	GST registration number

2.4 Inventory

Attribute	Data Type	Key	Description
Inventory_ID	INT	PK	Unique inventory ID
Product_ID	INT	FK	References Product
Quantity	INT	-	Available quantity
Minimum_Stock	INT	-	Minimum stock level
Warehouse	VARCHAR(100)	-	Warehouse location
Last_Updated	DATE	-	Last updated date

2.5 Orders

Attribute	Data Type	Key	Description
Order_ID	INT	PK	Unique order ID
Customer_ID	INT	FK	References Customer
Order_Date	DATE	-	Order date
Total_Amount	DECIMAL(10,2)	-	Total amount
Order_Status	VARCHAR(30)	-	Pending/Shipped/Delivered
Shipping_Address	TEXT	-	Delivery address
Delivery_Date	DATE	-	Expected delivery date

2.6 Order Details

Attribute	Data Type	Key	Description
Order_Detail_ID	INT	PK	Unique order detail ID
Order_ID	INT	FK	References Orders
Product_ID	INT	FK	References Product
Quantity	INT	-	Ordered quantity
Unit_Price	DECIMAL(10,2)	-	Price per product
Discount	DECIMAL(10,2)	-	Discount amount
Total_Price	DECIMAL(10,2)	-	Total price

2.7 Payment

Attribute	Data Type	Key	Description
Payment_ID	INT	PK	Unique payment ID
Order_ID	INT	FK	References Orders
Payment_Method	VARCHAR(30)	-	UPI/Card/COD
Transaction_ID	VARCHAR(100)	-	Transaction reference
Payment_Status	VARCHAR(20)	-	Success/Pending
Payment_Date	DATE	-	Payment date
Amount	DECIMAL(10,2)	-	Paid amount

2.8 Admin

Attribute	Data Type	Key	Description
Admin_ID	INT	PK	Unique admin ID
Admin_Name	VARCHAR(100)	-	Administrator name
Email	VARCHAR(100)	-	Email address
Phone	VARCHAR(15)	-	Contact number
Role	VARCHAR(30)	-	Admin role
Username	VARCHAR(50)	-	Login username
Password	VARCHAR(100)	-	Login password

2.9 Relationships

Entity 1	Relationship	Entity 2	Cardinality
Customer	Places	Orders	1 : M
Orders	Contains	Order Details	1 : M
Product	Appears In	Order Details	1 : M
Category	Contains	Product	1 : M
Supplier	Supplies	Product	1 : M
Product	Stored In	Inventory	1 : 1
Orders	Has	Payment	1 : 1
Admin	Manages	Product	1 : M
Admin	Manages	Orders	1 : M

3.0 Review

Attribute	Data Type	Key
Review_ID	NUMBER(10)	PK
Customer_ID	NUMBER(10)	FK
Product_ID	NUMBER(10)	FK
Review_Text	VARCHAR2(500)	—
Review_Date	DATE	—

3.1 Rating

Attribute	Data Type	Key
Rating_ID	NUMBER(10)	PK
Customer_ID	NUMBER(10)	FK
Product_ID	NUMBER(10)	FK
Rating_Value	NUMBER(1)	—
Rating_Date	DATE	—

3.2 Delivery

Attribute	Data Type	Key
Delivery_ID	NUMBER(10)	PK
Order_ID	NUMBER(10)	FK
Delivery_Date	DATE	—
Delivery_Status	VARCHAR2(20)	—
Delivery_Address	VARCHAR2(200)	—
Delivery_Partner	VARCHAR2(100)	—